Spectral distribution of synchrotron radition

Periodic motion with frequency $\omega_B = \frac{qB}{m\gamma}$.

Non-relativistically, this electric dipole radiation $\Rightarrow$ emitted waves have frequency ω.

<u>Relativistically</u>, there are higher harmonics $n\omega_B$!

Expand vector potential in series

$$\vec{A}(t) = \sum_n \vec{A}_n e^{-in\omega t} \qquad n \neq 0.$$

$$\vec{A}_n = \frac{1}{T}\int_0^T dt\, e^{in\omega t}\, \vec{A}(t)$$

$$\vec{A}_n = \frac{\mu_0 q}{4\pi R T}\int_0^T \frac{dt}{1-\frac{\hat{n}\cdot\vec{v}(\tilde{t})}{c}}\, \vec{v}(\tilde{t})\, e^{in\omega t} \quad = \left(dt = \frac{dt}{d\tilde{t}}d\tilde{t}\right)$$

$$dt = \left(1-\frac{\hat{n}\cdot\vec{v}}{c}\right)d\tilde{t} \quad , t = \tilde{t} + \frac{|\vec{R}-\vec{r}(\tilde{t})|}{c}$$

$$\vec{A}_n = \frac{\mu_0 q}{4\pi R T}\int_0^T d\tilde{t}\, \vec{v}(\tilde{t})\, e^{in\omega\left(\tilde{t} + |\vec{R}-\vec{r}(\tilde{t})|/c\right)}$$

$$|\vec{R}-\vec{r}| = \sqrt{(\vec{R}-\vec{r})\cdot(\vec{R}-\vec{r})} = \sqrt{R^2 - 2\vec{R}\cdot\vec{r} + r^2} =$$

$$= R\sqrt{1 - \frac{2\hat{R}\cdot\vec{r}}{R} + \frac{r^2}{R^2}} \approx R\left(1 - \frac{\hat{R}\cdot\vec{r}}{R}\right) = R - \hat{n}\cdot\vec{r}$$

$$\vec{A}_n = \frac{\mu_0 q}{4\pi R T}\int_0^T d\tilde{t}\, \vec{v}(\tilde{t})\, e^{in\omega\tilde{t} + in\frac{\omega}{c}(R-\hat{n}\cdot\vec{r}(\tilde{t}))} =$$

$$= \frac{\mu_0 q}{4\pi R T}\, e^{in\omega R/c}\int_0^T d\tilde{t}\, \vec{v}(\tilde{t})\, e^{in\omega\left(\tilde{t} - \frac{\hat{n}\cdot\vec{r}(\tilde{t})}{c}\right)}$$

Note: The period of motion is the same for t and $\tilde{t}$! $t \to t+T$ do not change $\tilde{t} \to \tilde{t}+T$ $\vec{r}(\tilde{t})$

Indeed, do $t \to t+T$, $\tilde{t} \to \tilde{t}+T$ in

$$c(t-\tilde{t}) = |\vec{R} - \vec{r}(\tilde{t})| \to c(t-\tilde{t}) = |\vec{R} - \vec{r}(\tilde{t}+T)|$$

$$\Rightarrow \vec{r}(\tilde{t}) = \vec{r}(\tilde{t}+T).$$

$$\vec{B}_n = \vec{\nabla}_R \times \vec{A}_n = \vec{\nabla}_R \times \frac{\mu_0 q}{4\pi R T} e^{ik_n \hat{n}\cdot\vec{R}} \int_0^T d\tilde{t}\, \vec{v}(\tilde{t})\, e^{in\omega\left(\tilde{t} - \frac{\hat{n}\cdot\vec{r}}{c}\right)}$$

$$k_n = \frac{n\omega}{c}$$

$$= i k_n \hat{n} \times A_n \quad (\text{comes from } e^{ik_n \hat{n}\cdot\vec{R}})$$

$\vec{E}_n = c\vec{B}_n \times \hat{n}$ (plane wave)

Consider next $\vec{A}_n^* = \left(\frac{1}{T}\int_0^T dt\, e^{in\omega t}\, \vec{A}(t)\right)^* =$

$$= \frac{1}{T}\int_0^T dt\, e^{-in\omega t}\, \underbrace{\vec{A}^*(t)}_{} = \frac{1}{T}\int_0^T dt\, e^{-in\omega t}\, \vec{A}(t) = \vec{A}_{-n}.$$

$\vec{A}(t)$ because $\vec{\nabla}_R \times \vec{A} = \vec{B}$ is observable.

Then Poynting vector

$$\overline{\vec{S}} = \frac{1}{\mu_0} \sum_{mm'} \vec{E}_m \times \vec{B}_{m'} \underbrace{\overline{e^{-i(m+m')t}}}_{\delta_{m+m',0}} = \frac{1}{\mu_0} \sum_m \vec{E}_m \times \vec{B}_{-m} =$$

$$= \frac{c}{\mu_0} \sum_m \left(\vec{B}_m \times \hat{n}\right) \times \vec{B}_{-m} = \hat{n}\, \frac{c}{\mu_0} \sum_m |\vec{B}_m|^2 =$$

$$= \hat{n}\, \frac{2c}{\mu_0} \sum_{m=1}^{\infty} |\vec{B}_m|^2.$$

Energy flux is represented by the sum of harmonics.

$$\left(\vec{A}_m\right)_x = \frac{\mu_0 q}{4\pi R T} e^{i k_m R} \int_0^T d\tilde{t} \; V_x(\tilde{t}) \; e^{i m \omega \left(\tilde{t} - \frac{\hat{n}\cdot\vec{r}(\tilde{t})}{c}\right)}$$

For synchrotron motion $\vec{V}(\tilde{t}) = v\left(\cos\omega\tilde{t}, \sin\omega\tilde{t}, 0\right)$

$$\vec{r}(\tilde{t}) = \vec{r}_0 + \frac{v}{\omega}\left(\sin\omega\tilde{t}, -\cos\omega\tilde{t}, 0\right)$$

$$\Rightarrow m\omega \frac{\vec{r}(\tilde{t})\cdot\hat{n}}{c} = \frac{m\omega}{c}\frac{v}{\omega}\left(\sin\omega\tilde{t}, -\cos\omega\tilde{t}, 0\right)\cdot\underbrace{\left(\sin\theta, 0, \cos\theta\right)}_{(n_\perp, 0, n_z)}$$

$$= m\beta \sin\omega\tilde{t}\, n_\perp$$

$$\left(\vec{A}_m\right)_x = \frac{\mu_0 q v}{4\pi R T} e^{i k_m R} \int_0^T d\tilde{t} \cos\omega\tilde{t} \; e^{i m\omega\tilde{t} - i m\beta n_\perp \sin\omega\tilde{t}}$$

$$= \frac{\mu_0 q v}{4\pi R} e^{i k_m R} \int_0^{2\pi} \frac{d\varphi}{2\pi} \cos\varphi \; e^{i m(\varphi - \beta n_\perp \sin\varphi)}$$

where $\varphi = \omega\tilde{t}$ and $\omega T = 2\pi$.

And $$\left(\vec{A}_m\right)_y = \frac{\mu_0 q v}{4\pi R} e^{i k_m R} \int_0^{2\pi} \frac{d\varphi}{2\pi} \sin\varphi \; e^{i m(\varphi - \beta n_\perp \sin\varphi)}$$

Bessel function of integer order $\underline{m}$

$$J_m(z) = \int_0^{2\pi} \frac{d\varphi}{2\pi} e^{i m\varphi - i z \sin\varphi}$$

and some identities

$$J_{m-1}(z) + J_{m+1}(z) = \frac{2m}{z} J_m(z); \quad J_{m-1}(z) - J_{m+1}(z) = 2 J_m'(z).$$

Comment on initial phase ψ which becomes important if there are several charges!

let $\vec{v}(\tilde{t}) = v\left(\cos(\omega\tilde{t}+\psi), \sin(\omega\tilde{t}+\psi), 0\right)$

$$\Rightarrow \vec{r}(\tilde{t}) = \vec{r}_0 + \frac{v}{\omega}\left(\sin(\tilde{\omega}t+\psi), -\cos(\tilde{\omega}t+\psi), 0\right)$$

$$\Rightarrow \frac{m\omega}{c}\vec{r}\cdot\hat{n} = \frac{m\omega}{\cancel{c}}\frac{\vec{r}_0\cdot n}{c} + \frac{m\omega}{c}\hat{n}\cdot\left(\sin(\tilde{\omega}t+\psi), -\cos(\tilde{\omega}t+\psi), 0\right)$$

$$(\vec{A}_m)_x = \frac{\mu_0 q v}{4\pi R T} e^{ikmR}\int_0^T d\tilde{t}\, \cos(\omega\tilde{t}+\psi) e^{im\omega\tilde{t} - im\omega\frac{\vec{r}_0\cdot\hat{n}}{c}} e^{-im\beta n_\perp \sin(\tilde{\omega}t+\psi)}$$

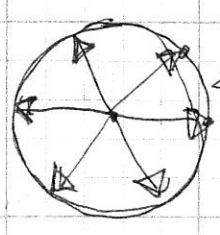
Now let $\varphi = \omega\tilde{t}+\psi$

$$T = \frac{2\pi}{\omega} \Rightarrow \frac{d\tilde{t}}{T} = \frac{d\varphi}{2\pi}$$

$$= \frac{\mu_0 q v}{4\pi R} e^{ikm(R-\vec{r}_0\cdot\hat{n})}\int_{\cancel{0}}^{2\pi+\psi}\frac{d\varphi}{2\pi}\cos(\varphi+\psi)\, e^{im(\omega\tilde{t}+\psi)} e^{-im\psi - im\beta n_\perp \sin(\varphi+\psi)}$$

$$(\varphi\to\varphi+\psi) \qquad = \frac{\mu_0 q v}{4\pi R} e^{ikm(R-\vec{r}_0\cdot\hat{n})}\int_0^{2\pi}\frac{d\varphi}{2\pi}\cos\varphi\, e^{im(\varphi-\beta n_\perp\sin\varphi-\psi)}$$

phase shift



← differ by phase difference

$$\vec{A}_x(t) = \sum_n A_{nx} e^{-in\omega t} = \frac{\mu_0 q v}{4\pi R}\sum_{\ell=1}^{N}\sum_m e^{ikmR} e^{-im\psi_\ell}$$

$$\cdot\int_0^{2\pi}\frac{d\varphi}{2\pi}\begin{pmatrix}\cos\varphi\\ \sin\varphi\end{pmatrix} e^{im\varphi - im\beta n_\perp\sin\varphi}$$

particles

harmonics

But $\int_0^{2\pi} \frac{d\varphi}{2\pi} \underbrace{\frac{e^{i\varphi}-e^{-i\varphi}}{2i}}_{\sin\varphi} e^{im\varphi - im\beta n_\perp \sin\varphi} =$

$$= \frac{J_{m+1}(m\beta n_\perp) - J_{m-1}(m\beta n_\perp)}{2i} = i\, J_m'(m\beta n_\perp)$$

where $J_m'(x) = \frac{d}{dx} J_m(x)$.

Also $\int \frac{d\varphi}{2\pi} \underbrace{\frac{e^{i\varphi}+e^{-i\varphi}}{2}}_{\cos\varphi} e^{im\varphi} e^{-im\beta n_\perp \sin\varphi} =$

$$= \frac{1}{2}\left(J_{m+1}(m\beta n_\perp) + J_{m-1}(m\beta n_\perp)\right) = \frac{1}{2}\frac{2m}{m\beta n_\perp} J_m(m\beta n_\perp)$$

$$= \frac{1}{\beta n_\perp} J_m(m\beta n_\perp).$$

$$\Rightarrow \vec{A}_m = \frac{\mu_0 q v}{4\pi} \frac{e^{ik_m R}}{R} \left(\frac{J_m(\beta m n_\perp)}{\beta n_\perp}, i J_m'(\beta n_\perp m), 0 \right)$$

$$\vec{B}_m = i k_n \hat{n} \times \vec{A}_m$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ n_\perp & 0 & n_z \\ A_{mx} & A_{my} & 0 \end{vmatrix} = -\hat{x}\, n_z A_{my} + \hat{y}\, n_z A_{mx} + \hat{z}\, n_\perp A_{my}$$

$$\vec{B}_m = i k_m \left\{ n_z (\vec{A}_m)_x \hat{y} - (\hat{x} n_z - \hat{z} n_\perp)(\vec{A}_m)_y \right\}$$

Note that vectors $\hat{y}$ and $(-n_z \hat{x} + n_\perp \hat{z})$ are orthogonal to each other, and $|\hat{y}| = 1$,

$$|-n_z \hat{x} + n_\perp \hat{z}| = n_z^2 + n_\perp^2 = 1$$

$$\Rightarrow |\vec{B}_m|^2 = k_m^2 |A_{mx}|^2 n_z^2 + k_m^2 |A_{my}|^2.$$

$$\overline{\frac{dP}{d\Omega}} = \overline{\vec{S}_0 \cdot \hat{n}}\, R^2 = \frac{2c}{\mu_0} R^2 \sum_{m=1}^{\infty} \left(\frac{m\omega}{c}\right)^2 \left(\frac{\mu_0 q v}{4\pi R}\right)^2 \left[\left(\frac{\cos\theta}{\beta \sin\theta} J_m(m\beta\sin\theta)\right)^2 + J_m'^2(m\beta\sin\theta)\right]$$

$n_\perp = \sin\theta$, $n_z = \cos\theta$

Individual terms in this sum represent contributions at frequency $m\omega$. $\Rightarrow$ spectral decomposition.

The sum over m can actually be carried over using identities

$$\sum_{m=1}^{\infty} m^2 J_m'^2(mz) = \frac{4+3z^2}{16(1-z^2)^{5/2}} \qquad z = \beta\sin\theta$$

$$\sum_{m=1}^{\infty} m^2 J_m^2(mz) = \frac{z^2(4+z^2)}{16(1-z^2)^{5/2}}$$

Doing so will just give us previously derived result for $\overline{\frac{dP}{d\Omega}}$.

It is more useful to analyze spectral content of $\overline{\frac{dP}{d\Omega}}$.
Check non-relativistic limit first:

$$\beta \to 0 \Rightarrow J_m(\beta m \sin\theta) \approx \frac{1}{m!}\left(\frac{\beta m \sin\theta}{2}\right)^m \Rightarrow J_m'(\beta m \sin\theta) \approx \frac{1}{(m-1)!}\left(\frac{\beta m \sin\theta}{2}\right)^{m-1}$$

$$\overline{\frac{dP}{d\Omega}} = \frac{c\mu_0 q^2 \omega^2 \beta^2}{8\pi^2} \sum_{m=1}^{\infty} m^2 \left\{ \frac{\cos^2\theta}{\beta^2 \sin^2\theta}\left[\left(\frac{\beta m \sin\theta}{2}\right)^m \frac{1}{m!}\right]^2 + \left[\frac{1}{(m-1)!}\left(\frac{\beta m \sin\theta}{2}\right)^{m-1}\right]^2 \right\}$$

Clearly for $\beta \to 0$ the main contribution is from $\underline{m=1}$ and it reproduces radiation of a spinning electric <u>dipole</u>

$$\frac{\mu_0 q^2 v^2 \omega^2}{32\pi^2 c}(1+\cos^2\theta).$$

Large n-limit

$$J_n(n\beta\sin\theta) = \int \frac{d\varphi}{2\pi}\, e^{in\varphi - in\beta\sin\theta\sin\varphi}$$

$\beta\sin\theta \sim 1$ for $\theta = \frac{\pi}{2} - \alpha$, $\alpha \ll 1$: $\beta\sin\theta = \beta\cos\alpha$

$\Rightarrow$ can expand: $n\varphi - n\beta\cos\alpha \underbrace{\sin\varphi}_{\varphi - \frac{1}{6}\varphi^3} = n\varphi(1-\beta\cos\alpha) + \frac{n\beta\cos\alpha}{6}\varphi^3$

$$= n\varphi\left(1-\beta+\tfrac{1}{2}\beta\alpha^2\right) + \frac{n\beta}{6}\left(1-\tfrac{1}{2}\alpha^2\right)\varphi^3$$

$$1-\beta \approx \frac{1}{2\gamma^2}$$

$$= \frac{1}{2}\, n\varphi\left(\frac{1}{\gamma^2}+\alpha^2\right) + \frac{n\varphi^3}{6}.$$

$$\Rightarrow J_n(n\beta\sin\theta) \approx \int_{-\infty}^{\infty} \frac{d\varphi}{2\pi}\, e^{i\left(\frac{1}{\gamma^2}+\alpha^2\right)\frac{n\varphi}{2} + i\frac{n}{6}\varphi^3} =$$

let $\frac{n}{6}\varphi^3 = \frac{1}{3}z^3$, $z = \left(\frac{n}{2}\right)^{1/3}\varphi \Rightarrow \frac{n\varphi}{2} = \frac{1}{2}\left(\frac{2}{n}\right)^{1/3} z\, n = \left(\frac{n}{2}\right)^{2/3} z$

$$= \left(\frac{2}{n}\right)^{1/3}\int_{-\infty}^{\infty}\frac{dz}{2\pi}\, e^{i\left(\frac{1}{\gamma^2}+\alpha^2\right)\left(\frac{n}{2}\right)^{2/3} z + i\frac{1}{3}z^3} = \frac{1}{\pi}\left(\frac{2}{n}\right)^{1/3}\int_0^{\infty}\cos\left[\left(\frac{n}{2\gamma^3}\right)^{2/3}(1+\gamma^2\alpha^2)z + \frac{1}{3}z^3\right]dz$$

$$= \left(\frac{2}{n}\right)^{1/3}\frac{1}{\sqrt{\pi}}\,\Phi\left[\left(\frac{n}{2}\right)^{2/3}\left(\frac{1}{\gamma^2}+\alpha^2\right)\right] = \left(\frac{2}{n}\right)^{1/3}\frac{1}{\sqrt{\pi}}\,\Phi\left[\left(\frac{n}{2\gamma^3}\right)^{3/2}(1+\gamma^2\alpha^2)\right]$$

Airy function

$$\Phi(t) \to \begin{cases} \frac{1}{2t^{1/4}}\, e^{-\frac{2}{3}t^{3/2}}, & t\to\infty \\ \frac{1}{(-t)^{1/4}}\sin\left(\frac{2}{3}(-t)^{3/2}+\frac{\pi}{4}\right), & t\to-\infty \end{cases}$$

$$\Phi(t) = \sqrt{\frac{t}{3\pi}}\, K_{1/3}\left(\frac{2}{3}t^{3/2}\right) \text{ and } K_\nu(u) \approx \sqrt{\frac{\pi}{2u}}\, e^{-u}$$

for $u\to\infty$

Similarly $J_n'(n\beta\sin\theta) = \left.\dfrac{dJ_n(z)}{dz}\right|_{z=n\beta\sin\theta} = -i\displaystyle\int_0^{2\pi}\frac{d\varphi}{2\pi}\sin\varphi\, e^{in\varphi - in\beta\sin\theta\sin\varphi}$

$$\approx -i\int_{-\infty}^{\infty}\frac{d\varphi}{2\pi}\,\varphi\, e^{i(1+\gamma^2\alpha^2)\frac{n\varphi}{2\gamma^2} + i\frac{n}{6}\varphi^3} = (-i)\left(\frac{2}{n}\right)^{2/3}\int_{-\infty}^{\infty}\frac{dz}{2\pi}\cdot z\cdot$$

$$\cdot\, e^{i(1+\gamma^2\alpha^2)\left(\frac{n}{2\gamma^3}\right)^{2/3} z + i\frac{z^3}{3}} =$$

$$= \frac{1}{\pi}\left(\frac{2}{n}\right)^{2/3}\int_0^{\infty} dz\; z\,\sin\left[\left(\frac{n}{2\gamma^3}\right)^{2/3}(1+\gamma^2\alpha^2)\,z + \frac{1}{3}z^3\right]$$

$$az + \frac{z^3}{3} = b\left(y + \frac{y^3}{3}\right) \Rightarrow \frac{z^3}{3az} = \frac{z^2}{3a} = \frac{by^3}{3yb} = \frac{y^2}{3}$$

$$\Rightarrow y = \frac{z}{\sqrt{a}}$$

$$z = \sqrt{a}\,y \Rightarrow a\sqrt{a}\,y + \frac{1}{3}(\sqrt{a})^3 y^3 = a^{3/2}\left(y + \frac{1}{3}y^3\right)$$

$$b = a^{3/2}$$

So let $z = \left(\frac{n}{2\gamma^3}\right)^{3/4}(1+\gamma^2\alpha^2)^{1/2}\, y$

$$\Rightarrow J_n'(n\beta\cos\alpha) = \frac{1}{\pi}\left(\frac{2}{n}\right)^{2/3}\left(\frac{n}{2\gamma^3}\right)^{3/2}(1+\gamma^2\alpha^2)\int_0^{\infty} dy\cdot$$

$$\cdot\, y\cdot \sin\left[\left(\frac{n}{2\gamma^3}\right)^{\frac{9}{4}}(1+\gamma^2\alpha^2)^{3/2}\left(y + \frac{y^3}{3}\right)\right]$$

But $\displaystyle\int_0^{\infty} dy\, y\,\sin\left[b\left(y + \frac{y^3}{3}\right)\right] = -\frac{1}{\sqrt{3}}K_{2/3}\left(\frac{2}{3}b\right).$

$$\Rightarrow J_n'(n\beta\cos\alpha) = -\frac{1}{\sqrt{3}\,\pi}\,\frac{1+\gamma^2\alpha^2}{\gamma^{9/2}}\,K_{2/3}\left(\frac{n}{2\gamma^3}\right)^{9/4}$$

Here $a = \left(\frac{n}{2\gamma^3}\right)^{2/3}(1+\gamma^2\alpha^2).$

$$z = \sqrt{a}\,y$$

$$J_n'(n\beta\cos\alpha) = \frac{1}{\pi}\left(\frac{2}{n}\right)^{2/3}\cdot a \underbrace{\int_0^\infty dy\, y \sin\left[a^{3/2}\left(y+\frac{y^3}{3}\right)\right]}_{-\frac{1}{\sqrt{3}} K_{2/3}\left(\frac{2}{3}a^{3/2}\right)} =$$

$$= -\frac{1}{\sqrt{3}\pi}\,\frac{1+\gamma^2\alpha^2}{\gamma^2}\,K_{2/3}\left[\frac{n}{3\gamma^3}\left(1+\gamma^2\alpha^2\right)^{3/2}\right].$$

And $$J_n(n\beta\cos\alpha) = \frac{1}{\pi}\left(\frac{2}{n}\right)^{1/3}\int_0^\infty dz\,\cos\left[\left(\frac{n}{2\gamma^3}\right)^{2/3}\left(1+\gamma^2\alpha^2\right)z+\frac{z^3}{3}\right]$$

$$= \frac{1}{\pi}\left(\frac{2}{n}\right)^{1/3}\left(1+\gamma^2\alpha^2\right)^{1/2}\left(\frac{n}{2\gamma^3}\right)^{1/3}\underbrace{\int_0^\infty dy\,\cos\left[a^{3/2}\left(y+\frac{y^3}{3}\right)\right]}_{\frac{1}{\sqrt{3}} K_{1/3}\left(\frac{2}{3}a^{3/2}\right)} =$$

$$= \frac{1}{\sqrt{3}\pi}\,\frac{\sqrt{1+\gamma^2\alpha^2}}{\gamma}\,K_{1/3}\left[\frac{n}{3\gamma^3}\left(1+\gamma^2\alpha^2\right)^{3/2}\right].$$

Since $K(u) \sim \sqrt{\frac{\pi}{2u}}\,e^{-u}$ for $u \gg 1$ we see that for $n \gg \gamma^3$ the contribution of $K_{1/3}$, $K_{2/3}$ becomes exponentially small $\Rightarrow$ typical $n \sim \gamma^3$.

$\Rightarrow$ typical frequency $\omega \sim \gamma^3\omega_B \gg \omega_B$!

$$\frac{\cos^2\theta}{\beta^2 \sin^2\theta} \to \frac{\sin^2\alpha}{\beta^2\cos^2\alpha} \approx \frac{\alpha^2}{1\cdot 1}$$

$$\gamma^2 = \frac{1}{1-\beta^2} \Rightarrow 1-\beta^2 = \frac{1}{\gamma^2}, \quad \beta^2 = 1 - \frac{1}{\gamma^2} \approx 1.$$

$$\overline{\frac{dP}{d\Omega}} = \frac{\mu q^2 \omega_B^2 v^2}{8\pi^2 c} \sum_{n=1}^{\infty} \frac{n^2}{3\pi^2} \left\{ \frac{\alpha^2(1+\gamma^2\alpha^2)}{\gamma^2} K_{1/3}^2\left[\frac{n}{3\gamma^3}(1+\gamma^2\alpha^2)^{3/2}\right] \right.$$

$$\left. + \left(\frac{1+\gamma^2\alpha^2}{\gamma^2}\right)^2 K_{2/3}^2\left[\frac{n}{3\gamma^3}(1+\gamma^2\alpha^2)^{3/2}\right] \right\} = \sum_{n=1}^{\infty} \overline{\frac{dP_n}{d\Omega}}$$

Here $\omega = \omega_B = \dfrac{qB}{\gamma m}$.

All frequencies $n\omega_B$ contribute, with $n \sim \gamma^3$, so the spectrum is very dense: $\omega = n\omega_B$

$$\Rightarrow \overline{\frac{d^2P}{d\omega\, d\Omega}} = \overline{\frac{dP_n}{d\Omega}}\, dn = \overline{\frac{dP_n}{d\Omega}} \frac{d\omega}{\omega_B}$$

and $\displaystyle\sum_n \overline{\frac{dP_n}{d\Omega}} \to \int \frac{d\omega}{\omega_B} \overline{\frac{dP_n}{d\Omega}}$

This can also be integrated over θ to obtain full intensity $\left(d\Omega = 2\pi \sin\theta\, d\theta \approx 2\pi\, d\alpha\right)$

The end result looks quite similar to eqns. (6.30a) + (6.30b) from the textbook.